

Process Kernel - Implementation Specification

What to build, what to enforce, and what must remain configurable in a buyout workflow platform.

One process kernel. Discrete authority gates. Continuous underwriting. Configurable execution.

How to use this specification

This document translates the operating logic of a buyout process into system objects, state transitions, rules, configuration flags, permissions, interfaces, and acceptance criteria.

1. Domain architecture

Fund policy, deal process, core entities, process spine.

2. Runtime engine

State machine, gates, loops, materiality, synchronization.

3. Workstream modules

Commercial, financial, operational, technology, talent, legal, tax, financing.

4. Underwriting state

Model maturity, assumptions, scenarios, value-creation maturity.

5. Decision controls

Open items, readiness, authority, permissions.

6. Rule system

AXIOM, ENFORCE, CONFIGURE, SUGGEST and the complete rule catalog.

7. Configuration

Deal archetypes, fund policy, capability routing, co-investment.

8. Technical implementation

Schema, events, APIs, integrations, UI, security, monitoring.

The platform must preserve one common economic and governance logic while allowing firms, funds, and transactions to express different operating models.

Fund logic becomes system policy

Fund-level logic

- Investment mandate: control, significant minority, sector, geography, check range, holding period.
- Portfolio constraints: concentration, reserves, follow-on capacity, currency, leverage and liquidity limits.
- Governance: who may advise, recommend, authorize spend, approve an offer, approve an investment, execute and fund.
- Operating model: internal specialist teams, external advisers, portfolio resources and committee structure.
- Ownership model: board rights, monitoring cadence, value-creation expectations and exit authority.

Technical representation

- `FundPolicy` supplies defaults and hard limits.
- `DealProcess` instantiates the common kernel for a specific opportunity.
- `ArchetypeFlag` activates conditional modules.
- `AuthorityPolicy` controls legal transitions and role permissions.
- `CapabilityRoute` assigns each required function internally or externally.

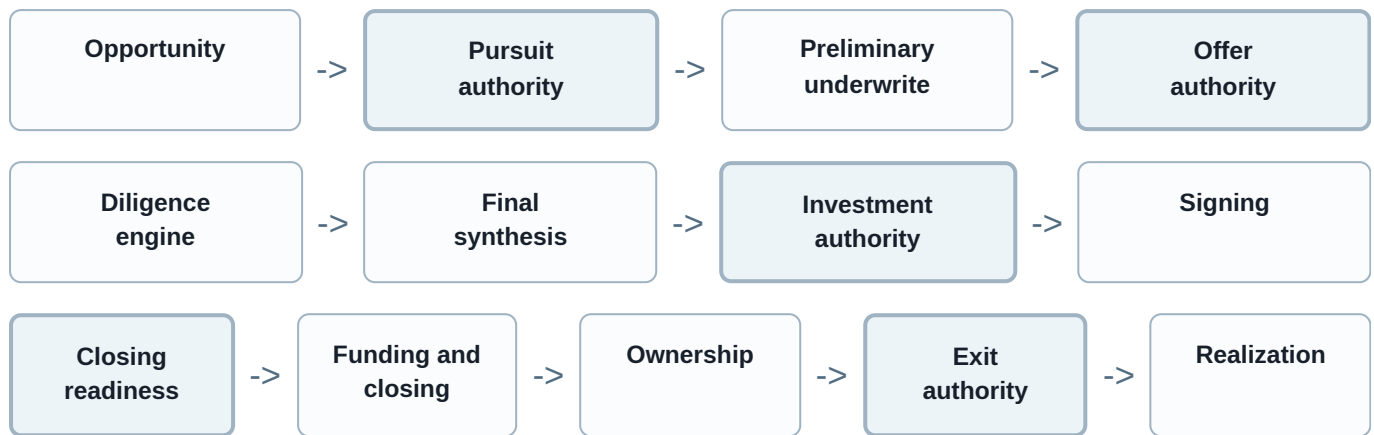
Do not create a separate workflow template for each fund. Create one kernel and let fund policy, deal archetype and capability routing determine how it runs.

Core domain objects

Object	Purpose	Minimum fields
FundPolicy	Defines mandate, limits, governance and default modules.	strategy; control_profile; limits; authority_policy_id; capability_profile_id
DealProcess	Owns the lifecycle of one opportunity.	fund_id; phase; status; archetype_flags; lead_owner; materiality_profile
Workstream	Tracks one diligence or execution function.	type; owner; scope; status; findings; open_item_ids; latest_update
ModelVersion	Stores the economic state used for decisions.	version; state; assumptions; scenarios; outputs; parent_version_id; approval_status
VCPInitiative	Stores a value-creation lever and its maturation.	lever; state; owner; value; cost; timing; dependencies; model_links
OpenItem	Represents unresolved decision-relevant uncertainty.	category; materiality; status; owner; due_event; resolution_path; gate_effect
DecisionRecord	Records recommendations and authorities.	authority_type; body; decision; conditions; rationale; scope; timestamp
ClosingCondition	Controls release of funds and closing.	condition; owner; satisfaction_method; waiver_authority; status
Artifact	References a memo, report, model, term sheet or execution document.	type; version; location; related_objects; effective_status
ProcessEvent	Provides an immutable event history.	event_type; actor; object_id; payload; timestamp; correlation_id

Artifacts support the process but do not define it. A report can be absent while the underlying function still exists, and a report can be present while the underlying issue remains unresolved.

The canonical lifecycle



Ordered spine

The spine protects authority, commitment and capital release. Its order matters because each gate changes what the firm is permitted to do.

Iterative interior

The diligence engine is not a straight sequence. Workstreams, assumptions, price, structure, risk and value creation revise one another until the case is sufficiently bounded.

The product needs both models at once: a state machine for authority and a cyclic graph for underwriting.

Process phases and legal transitions

Phase	Entry condition	Exit condition	Permitted terminal outcomes
ORIGINATED	Opportunity record created.	Basic fit and ownership assigned.	Reject; park; advance to screen.
SCREENING	Minimum company and transaction data available.	Initial thesis, rough economics and key risks recorded.	Reject; request information; seek pursuit authority.
PURSUIT_AUTHORIZED	Authorized body permits resources and deeper work.	Workplan, budget and initial responsibilities active.	Pause; withdraw; build preliminary underwrite.
PRELIMINARY_UNDERWRITE	Sufficient data for price range, leverage and return cases.	Offer decision and diligence architecture prepared.	Reject; revise; authorize offer; proceed without offer where applicable.
DILIGENCE_ACTIVE	Access and workstreams are active.	Material issues have explicit dispositions and the decision case is coherent.	Withdraw; renegotiate; move to final synthesis.
FINAL_SYNTHESIS_READY	Economics, risks, terms, financing and plan are synchronized.	Valid investment decision recorded.	Reject; approve; approve with conditions; defer.
SIGNED	Binding commitment executed.	Closing conditions satisfied or validly waived.	Terminate; extend; close.
CLOSING_READY	Release controls completed.	Funds released and closing confirmed.	Hold; waive; fund.
OWNERSHIP	Investment closed.	Exit decision and execution readiness.	Hold; refinance; add-on; partial realization; exit.
REALIZED	Disposition completed.	Final proceeds and close-out recorded.	Archive.

Discrete decisions inside a continuous process

Pursuit gate

Purpose: authorize effort, access and cost.

- Minimum thesis and mandate fit
- Rough value and return range
- Known red flags and conflicts
- Proposed workplan, budget and owner
- Information-value case for deeper work

Investment gate

Purpose: authorize the binding economic decision.

- Final underwrite and downside
- Risk and open-item dispositions
- Financing and liquidity
- Legal, tax and structural terms
- Value-creation plan and ownership readiness

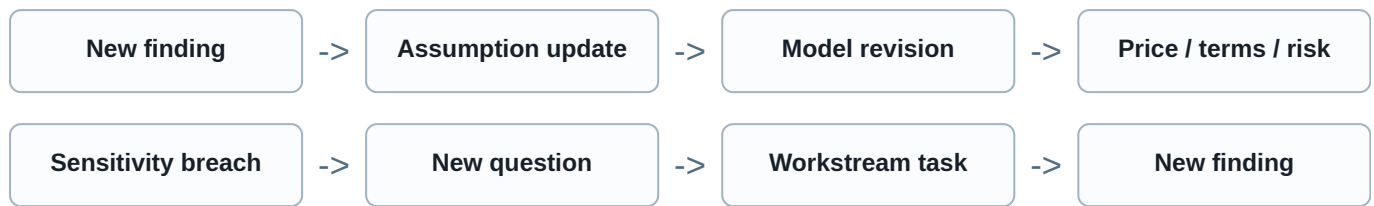
Closing gate

Purpose: authorize capital release.

- Conditions precedent
- Signed authority and execution documents
- Compliance and KYC/AML
- Financing and funds flow
- No unresolved material change

A gate is a decision event, not a document. The system may require a memo, but the controlling object is the authorized decision with scope, conditions and validity.

The diligence engine is cyclic



Static objects

- Fund policy
- Authority policy
- Workstream definitions
- Archetype conditions
- Field and enum definitions

Dynamic objects

- Assumptions and scenarios
- Model and memo versions
- Open-item states
- Price and structure
- Workstream scope and priorities

Do not implement a single “diligence complete” flag. Completion is local to a workstream; readiness is determined by the disposition of material open items and the coherence of the decision case.

Continuous loops and discrete checkpoints

Concept	Meaning in the investment process	Technical behavior
Continuous loop	Information and judgment evolve throughout the process.	Allow updates at any time; publish events; recalculate dependent views.
Discrete checkpoint	A body authorizes a bounded action at a point in time.	Create immutable decision records and transition guards.
Local closure	A workstream has completed its current scope.	Set workstream status to complete-for-scope, not globally final.
Material reopening	New information changes a previously accepted conclusion.	Reopen affected items, invalidate stale readiness and require re-approval when thresholds are crossed.
Dynamic priority	A sensitivity or risk becomes more decision-relevant.	Re-rank tasks and escalate owners without changing the canonical spine.
Convergence	The case is sufficiently bounded for a decision, not perfectly known.	Compute readiness from required outputs, material open-item disposition and authority conditions.

The engine should support both synchronous review meetings and asynchronous work. A meeting is one event source; it is not the workflow itself.

Seven required functions, flexible delivery

Workstream	Primary system question
Commercial	Are revenue, demand, pricing, retention and competitive assumptions credible?
Financial and accounting	What are normalized earnings, cash conversion, debt-like items and reliable historicals?
Operational	Can the business deliver the plan, at what cost, capacity and execution risk?
Talent and management	Can the leadership team execute, and what changes are required?
Legal and regulatory	Can the transaction be completed and risks allocated within acceptable bounds?
Tax and structure	How should ownership, financing and cash flows be structured?
Financing and capital markets	How much debt and liquidity can the business support under downside conditions?

Common contract

- Every workstream has a scope, owner, tasks, findings, assumptions affected, open items and gate impact.
- Every material finding must identify what changes: model, price, structure, contract, financing, plan or decision.
- Every workstream may be delivered internally, externally or through a hybrid team.
- The same function remains required even when the artifact or provider changes.
- Specialist submodules can be activated without creating a new process spine.

Turn market claims into operating assumptions

What it means

- Define the actual market and addressable profit pool.
- Test customer behavior, retention, churn, concentration and willingness to pay.
- Understand competition, substitution, channels, win/loss patterns and barriers.
- Translate pipeline and sales claims into probability-weighted conversion.

Why it matters

- Revenue is often the largest driver of value and the least directly controlled variable.
- Small errors in retention, pricing or volume compound through margins, leverage and exit value.
- Commercial weakness can invalidate an otherwise attractive operating plan.
- Commercial strength may justify capacity, hiring or acquisition investment.

Technical module

Component	Required behavior
Market model	Store market definitions, growth cases, cyclicalities and addressable segments separately from company forecasts.
Customer analysis	Support cohort, concentration, retention, churn, pricing and survey/interview findings.
Pipeline model	Separate booked, contracted, qualified, proposed and speculative opportunities.
Competition map	Track competitors, substitutes, relative positioning and expected response.
Assumption links	Map each commercial conclusion to revenue, margin, timing, terminal growth and exit assumptions.
Variation	Enable lighter internal work for simple markets and specialist modules for regulated, technical or data-rich markets.

Financial and accounting verification

The purpose is not to reproduce the accounts. It is to determine which earnings and cash flows are economically repeatable, which obligations belong to the buyer, and which figures can support price and leverage.

Core sequence

- Reconcile audited accounts, management accounts and transaction data.
- Normalize revenue and expenses for recurring economics.
- Assess working capital, cash conversion and seasonality.
- Identify capex, cash tax, debt and debt-like obligations.
- Test forecast accuracy and the reliability of management reporting.

System outputs

- Normalized EBITDA and bridge to reported EBITDA
- Free cash flow and conversion ratios
- Working-capital peg and purchase-price mechanics
- Net debt and debt-like schedule
- Forecast reliability score and control gaps
- Model adjustments with owner and rationale

A financial report is not the required object. The required object is a resolved economic bridge from reported performance to normalized earnings, cash and obligations.

EBITDA normalization - the operating algorithm

1. Establish the reported baseline

Reconcile statutory, management and transaction-period figures. Differences must be explained before adjustments are considered.

2. Define the perimeter and period

Confirm entities, sites, discontinued operations, acquisitions, disposals, accounting standards and the exact trailing period.

3. Classify each adjustment

Use controlled categories: non-recurring, non-operating, owner-specific, run-rate, pro forma, accounting reclassification, or unsupported.

4. Test recurrence and causality

Ask whether the cost or benefit is likely to recur, whether it is necessary to generate revenue, and whether the buyer will inherit it.

5. Test amount and timing

Verify source data, period allocation, partial-year effects, implementation dependencies and double counting.

6. Separate base from upside

Current recurring economics belong in normalized EBITDA. Future initiatives belong in the value-creation plan unless already implemented and demonstrably run-rate.

7. Propagate downstream

Approved adjustments update leverage, purchase price, covenants, cash flow, returns, sensitivities and negotiation positions.

EBITDA adjustment object model

Field	Purpose	Validation rule
adjustment_type	Controlled economic category.	Required; cannot be free text only.
reported_amount	Amount in the source period.	Must reconcile to a traceable account or schedule.
normalized_amount	Amount accepted into normalized EBITDA.	May differ from the proposed amount; requires rationale.
recurrence_assessment	Recurring, non-recurring, transitional or uncertain.	Uncertain items remain open or enter a scenario, not the base case.
buyer_inheritance	Whether the buyer inherits the economics.	Buyer-required costs cannot be removed solely because the seller did not incur them.
implementation_status	Not started, underway, complete, verified run-rate.	Future savings cannot enter current EBITDA without an explicit policy.
double_count_group	Prevents overlap with other adjustments or VCP initiatives.	Only one active item per economic effect unless allocation is explicit.
model_links	Assumptions and outputs affected.	Must include leverage and return links when material.

The platform should display a bridge from reported EBITDA to normalized EBITDA and a separate bridge from normalized EBITDA to potential future EBITDA.

Working capital, cash conversion and purchase price

Working-capital logic

- Define operating current assets and liabilities.
- Remove cash, debt, tax, transaction and non-operating balances.
- Analyze monthly seasonality, growth and one-off distortions.
- Calculate normalized levels using the correct business driver, not a single generic average.
- Set the peg and define the closing true-up mechanics.

Why it matters

- A low closing balance can transfer value from buyer to seller.
- Rapid growth can consume cash even when EBITDA rises.
- Seasonal businesses can look over- or under-funded depending on the closing month.
- Collections, inventory and supplier terms may reveal operational risk hidden in EBITDA.

Required system calculations

Calculation	Implementation
Monthly working capital	Store components by month, entity and business line.
Days metrics	DSO, DIO and DPO with consistent denominators and exception handling.
Normalized peg	Support average, seasonal curve, growth-adjusted and negotiated methods.
Cash conversion	EBITDA -> operating cash -> free cash flow bridge.
Closing true-up	Calculate delivered working capital, permitted exclusions and purchase-price adjustment.
Sensitivity	Show cash and equity impact for alternative peg and seasonality assumptions.

Can the business deliver the plan?

What it covers

- Capacity, utilization, throughput, yield and service delivery.
- Labor productivity, procurement, suppliers and supply chain.
- Quality, safety, inventory, facilities and maintenance.
- Capex, scalability, cost structure and operational resilience.
- Carve-out, integration, separation and implementation requirements.

Why firms differ

- Some firms maintain permanent operating teams; others route the same work through deal professionals or advisers.
- Asset-light services require different modules from manufacturing, healthcare or logistics.
- A platform acquisition needs a different depth of operating work than a small add-on.
- A stable business may need validation; a transformation thesis needs a fully costed execution plan.

Technical behavior

Object	Required fields
Operational baseline	KPI definition; source; historical range; current level; target; owner.
Constraint	Type; affected output; severity; capacity date; mitigation; capex.
Initiative	Benefit; cost; timing; owner; dependency; confidence; VCP link.
Site assessment	Site; process; capacity; quality; safety; systems; actions.
Implementation plan	Milestone; responsible role; budget; prerequisite; status.

Technology, data and cyber as operating dependencies

Technology is a specialist submodule that may sit inside operational, legal or standalone diligence depending on the business. It should never be treated as a checklist detached from the investment thesis.

Questions

- Can the architecture support growth, integration and reporting?
- Which systems are critical, obsolete, unsupported or dependent on individuals?
- What data is available, governed and reliable?
- What cyber, privacy and resilience risks could interrupt operations or create liability?
- What separation, migration or investment is required after closing?

System outputs

- Application and infrastructure map
- Critical dependency register
- Cyber and privacy issue list
- Data quality and analytics assessment
- Technology capex and implementation roadmap
- TSA, migration and integration requirements

Technology findings must link to cost, timing, operational continuity, legal exposure, value-creation initiatives and closing conditions.

Management capability and alignment

Operating questions

- Can the team deliver the underwritten plan?
- Where is the business dependent on a founder or a small number of people?
- Are finance, reporting and control functions strong enough for institutional ownership?
- Which roles must be retained, replaced or added?
- Are incentives, rollover and governance aligned with the plan?

Why it changes the deal

- A management gap changes timing, cost and the credibility of value creation.
- Founder dependence can affect structure, rollover, governance and transition arrangements.
- A weak finance function can reduce confidence in every other workstream.
- Leadership constraints can turn an attractive strategy into an execution risk.

Technical module

Object	Fields and behavior
Role assessment	Role; incumbent; criticality; performance; retention risk; action.
Succession item	Trigger; interim plan; candidate; timing; gate effect.
Incentive plan	Participants; instrument; vesting; performance conditions; rollover.
Governance design	Board roles; reserved matters; reporting cadence; escalation.
Hiring initiative	Role; budget; search owner; target date; dependency.

Legal and regulatory control plane

Core functions

- Confirm title, perimeter and authority to sell.
- Review material contracts, change of control and consents.
- Assess litigation, licenses, regulation, employment, pensions, environment, data and IP.
- Translate risk into conditions, covenants, indemnities, escrow, insurance and price.
- Control signing, closing and the satisfaction or waiver of conditions.

Technical consequences

- Legal issues can create hard blockers, conditional blockers or post-close actions.
- Contractual mitigation reduces exposure but does not erase residual counterparty or enforceability risk.
- The legal perimeter must synchronize with the model, tax structure and funds flow.
- Every waiver requires the correct authority and a durable decision record.

Objects

Object	Key fields
Legal issue	Category; document; obligation; exposure; mitigation; residual risk; owner.
Consent	Counterparty; requirement; deadline; status; consequence of failure.
Protection	Type; beneficiary; cap; duration; security; claims process.
Closing condition	Condition; satisfaction method; waiver authority; status.
Execution document	Version; approval status; signatories; effective state.

Tax and transaction structure

Tax and structure determine how the buyer owns the asset, finances it, moves cash, incentivizes management and ultimately exits. The module must connect structure to economics rather than treating tax as a standalone report.

Key design questions

- Asset or share acquisition?
- Which acquisition vehicles and jurisdictions?
- How is debt introduced and serviced?
- What withholding, transfer tax and interest limitations apply?
- How are management rollover and incentives structured?
- What historic liabilities, losses and exit taxes matter?

Required links

- Sources and uses
- Cash tax forecast
- Debt schedule and deductibility
- Purchase-price allocation
- Management economics
- Funds flow and closing steps
- Exit proceeds and tax leakage

The platform should version the legal entity structure and funds flow together. A structure change is a model event, a legal event and often an authority event.

Debt capacity, liquidity and certainty of funds

Core questions

- How much debt can the business support in base and downside cases?
- What are pricing, fees, amortization, maturity and covenant terms?
- How much liquidity is needed for seasonality, capex, acquisitions and shocks?
- What conditions could prevent funding?
- How does the financing package affect price, returns and strategic flexibility?

Why firms differ

- Internal capital-markets teams may run lender strategy and documentation.
- Smaller firms may rely on the deal lead, advisers or direct lenders.
- Unitranche, syndicated, bridge and asset-backed structures require different modules.
- Buy-and-build plans need acquisition capacity, not only closing leverage.

Technical outputs

Output	Required behavior
Debt schedule	Tranche, rate, fees, amortization, maturity, cash sweep, prepayment.
Covenant model	Definition, test dates, headroom, cure rights and downside breaches.
Liquidity view	Minimum cash, revolver, seasonal need, capex, working capital and contingency.
Lender process	Counterparty, stage, terms, conditions, probability and ownership.
Funds certainty	Committed amount, conditions, expiry, documentary status and release authority.

One change can affect the entire transaction

The platform must propagate material changes rather than storing isolated workstream conclusions.

Change	Immediate updates	Potential gate effect
Revenue forecast reduced	Model, leverage, price ceiling, covenant headroom, VCP priorities.	Re-approval if returns or authority limits are breached.
EBITDA adjustment rejected	Normalized earnings, debt capacity, purchase price and returns.	Offer or investment authority may become stale.
Capacity constraint identified	Capex, production plan, revenue timing, implementation plan.	Final approval may require a costed solution.
Management gap identified	Hiring cost, timing, governance, retention and VCP confidence.	Condition, post-close action or price adjustment.
Consent risk escalates	Legal protections, timing, financing conditions and closing plan.	Signing or closing may be blocked.
Tax structure changes	Cash tax, funds flow, debt deductibility and management economics.	Authority may need refresh if economics change materially.
Financing terms worsen	Equity requirement, returns, liquidity and acquisition capacity.	Price, structure or final approval may need revision.

Use dependency links and event-driven recalculation. Do not rely on users to remember every downstream effect manually.

M0-M6 underwriting maturity

State	Purpose	Minimum contents	Transition
M0 - Quick screen	Reject or prioritize.	Basic historicals, simple valuation, rough leverage and return range.	Pursuit authority and better data.
M1 - Preliminary underwrite	Set price range, risk agenda and initial cases.	Historical bridge, preliminary normalization, drivers, debt capacity, scenarios.	Offer position and full diligence.
M2 - Diligence operating model	Represent operating mechanics and questions.	Granular revenue, margins, working capital, capex, initiatives and management plan.	Findings resolve or redefine assumptions.
M3 - Diligence-adjusted case	Incorporate workstream conclusions.	Normalized earnings, revised drivers, bounded VCP, risk adjustments.	Price, structure and financing become actionable.
M4 - Financing and bid model	Support terms, financing and maximum price.	Sources and uses, debt, covenants, return cases, price sensitivities.	Final terms and open items are bounded.
M5 - Final underwrite	Support the binding investment decision.	Final cases, price, financing, VCP, risks, conditions and returns.	Valid authority and commitment.
M6 - Ownership budget	Translate the thesis into accountability.	Budget, KPIs, cash, leverage, milestones, hiring and reporting.	Ownership monitoring and reforecasting.

M0-M2 - from screen to operating model

M0

Optimize for speed.

- Few assumptions
- Wide ranges
- Explicit unknowns
- Fast rejection
- No false precision

M1

Optimize for decision framing.

- Price range
- Initial cases
- Debt capacity
- Risk agenda
- Workplan

M2

Optimize for causal detail.

- Operating drivers
- Cohorts or volume/price
- Cash mechanics
- Initiatives
- Sensitivity links

Technical rules

- Each state is a model purpose, not merely a version number.
- A model may contain multiple saved versions inside one state.
- The state changes only when the model is fit for the next decision purpose.
- Unknowns must be explicit assumptions or open items, never silent blanks.
- Every transition stores a parent version and a change summary.

M3-M5 - from findings to authority

● M3 - diligence-adjusted case

Replace preliminary assumptions with workstream conclusions. Separate accepted base economics from uncertain or contingent outcomes. Reconcile all major bridges.

● M4 - financing and bid model

Make price, structure, debt, liquidity, covenant and return trade-offs explicit. Support maximum-price and term comparisons.

● M5 - final underwrite

Freeze the decision version presented for authority. Record its assumptions, scenarios, open-item dispositions, conditions and model checksum.

A later model version must not silently replace the version that was approved. Material post-approval changes require a new decision or a formally permitted update path.

Decision-version controls

- Immutable approved snapshot
- Hash or checksum
- Linked decision record
- Materiality comparison against the approved case
- Re-approval threshold policy
- Clear distinction between approved, current and working versions

M6 - ownership budget and monitoring state

What changes at closing

- Investment assumptions become operating commitments.
- Annual figures become monthly or quarterly budgets.
- VCP initiatives receive owners, milestones and budgets.
- Leverage and liquidity become monitored constraints.
- Board and management reporting become recurring workflows.

System design

- Create M6 from the approved M5 snapshot, not from a fresh blank model.
- Preserve a bridge from underwrite to budget.
- Track variance by operational cause, not only accounting line.
- Allow reforecasting while retaining the original underwrite.
- Link every major initiative to KPIs, cash impact and milestones.

Core views

View	Purpose
Underwrite-to-budget bridge	Shows what changed between the acquisition case and the operating budget.
KPI dashboard	Tracks operational and financial drivers.
Liquidity and leverage	Monitors headroom and capital allocation.
Initiative delivery	Tracks VCP milestones, value, cost and risk.
Board pack status	Controls recurring governance outputs.

Assumption and scenario governance

Object	Required fields	Control
Assumption	Name; unit; value; effective period; owner; rationale; source object.	Versioned; dependency-linked; cannot be overwritten in place.
Scenario	Base, downside, upside, lender, management or custom.	Scenario inherits from a parent and stores only deltas.
Sensitivity	Input range; output metrics; threshold.	May generate tasks or reopen items when a threshold is crossed.
Bridge	From-version; to-version; categorized changes.	Required for material model transitions and re-approval.
Override	Field; old value; new value; actor; reason; authority.	Logged and visible in decision views.
Calculation policy	Normalization, cash conversion, leverage and return conventions.	Fund-level default with controlled transaction override.

Scenario design should separate uncertainty from aspiration. A value-creation lever is not automatically a base-case assumption.

V0-V5 value-creation maturity

State	Meaning	System requirement
V0 - Initial hypotheses	Potential levers derived from the thesis and sector view.	Capture lever, mechanism, affected KPI and initial questions.
V1 - Diligence validated	Levers retained, rejected or refined through workstream findings.	Record validation status, constraints and dependencies.
V2 - Quantified	Value, cost, timing, resources and cash effects estimated.	Link to model assumptions and prevent double counting.
V3 - Base vs upside	Levers allocated across downside, base and upside cases.	Require explicit scenario assignment and rationale.
V4 - Implementation ready	Owners, milestones, budgets, hires and first actions defined.	Create executable initiative plans.
V5 - Ownership and exit roadmap	Initiatives integrated into monitoring, capital allocation and exit preparation.	Track realized value and remaining opportunity.

V0-V2 - from idea to quantified initiative

V0 - hypothesis

State the mechanism, not only the label. “Pricing” is incomplete; “segment-specific price architecture raises realized price without unacceptable churn” is testable.

V1 - validation

Use commercial, operational, talent, financial, legal and technology findings to retain, reject or modify the lever.

V2 - quantification

Estimate EBITDA, cash, capex, one-time cost, timing, resources and dependencies. Keep ranges where precision is not justified.

VCP initiative schema

Field	Meaning
mechanism	Causal explanation of how the lever creates value.
baseline	Current KPI or economic state.
target	Expected future KPI or state.
gross_value	Economic benefit before implementation costs.
cash_cost	One-time and recurring cash requirements.
timing_curve	When costs and benefits occur.
dependencies	People, systems, capex, approvals, acquisitions or market conditions.
model_links	Assumptions and outputs affected.

V3-V5 - from underwriting to execution

V3

Allocate the lever.

- Downside
- Base
- Upside
- Excluded from model

V4

Make it executable.

- Owner
- Milestones
- Budget
- First 100/180 days
- Risk controls

V5

Manage through ownership.

- KPI monitoring
- Capital allocation
- M&A
- Refinancing
- Exit readiness

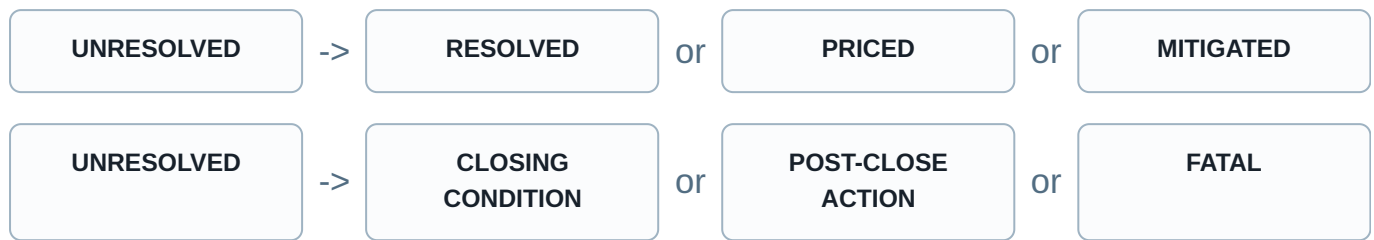
Control logic

- Do not count the same value in both the base model and a separate initiative.
- Do not accept a benefit without its cost, timing and dependency profile.
- Demote a lever when validation fails or implementation conditions change.
- Promote a lever only when its mechanism and inputs are sufficiently defined.
- At ownership, track realized value separately from forecast value.

A taxonomy for decision-relevant uncertainty

Status	Meaning	Runtime effect
UNRESOLVED	Material uncertainty remains without sufficient resolution or accepted treatment.	May block a gate depending on materiality and policy.
MATERIALLY_RESOLVED	Uncertainty is reduced below the decision threshold.	Close or monitor; retain the resolution record.
ACCEPTED_AND_PRICED	Residual risk is reflected in price, model, reserve or return.	Requires linked economic adjustment and authority.
CONTRACTUALLY_MITIGATED	Risk is allocated through contract, insurance, escrow, covenant or consent.	Track residual enforceability and counterparty risk.
CLOSING_CONDITION	Issue may remain at signing but must be satisfied or waived before funding.	Creates a closing-control object.
POST_CLOSE_ACTION	Risk is accepted because remediation can occur under ownership.	Requires owner, cost, timing, KPI and budget.
FATAL	Risk invalidates legality, financing, executability, mandate or economics.	Terminate or pause until facts fundamentally change.

Transitions, reopening and escalation



Transition rules

- Every transition requires a reason, responsible actor, supporting object and effective scope.
- New information may reopen any non-fatal item to UNRESOLVED.
- A contractual protection may fail and become unresolved or fatal.
- A closing condition becomes satisfied, waived, extended or failed; it does not become “complete” without an outcome.
- A post-close action must migrate into the ownership plan at closing.
- Materiality is dynamic: an item can become more important when price, leverage, timing or structure changes.

The open-item register is the control surface for convergence. It replaces vague statements that diligence is “done.”

Gate readiness is computed, not guessed

Gate	Required readiness checks
Pursuit authority	Mandate fit; owner; preliminary economics; red flags; proposed scope; budget; conflicts.
Offer authority	Price range; preliminary model; key assumptions; process terms; downside; authority limits.
Investment authority	Decision model; workstream dispositions; risk register; financing; legal/tax structure; VCP; open items.
Signing	Valid investment decision; approved documents; execution authority; signing conditions.
Closing readiness	Conditions precedent; compliance; financing; funds flow; signatures; no material adverse change.
Exit authority	Exit economics; buyer and execution plan; tax; debt; warranties; approvals.

Readiness result

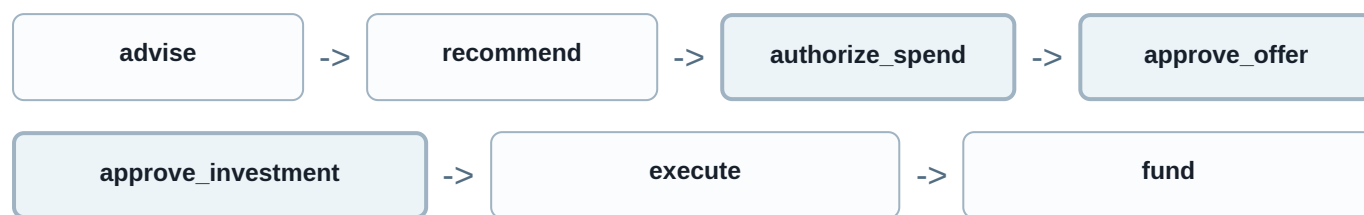
READY
 All mandatory conditions satisfied.

READY_WITH_CONDITIONS
 Permitted conditions are explicit and owned.

NOT_READY
 One or more blocking requirements fail.

STALE
 A material change invalidated prior readiness.

A richer authority model



Authority type	What it permits	What must be stored
advise	Provide specialist input.	Scope, advice, recipient and timestamp.
recommend	Propose an action to an authority body.	Recommendation, conditions and dissent.
authorize_spend	Commit diligence resources or cost.	Budget, limits, workplan and expiry.
approve_offer	Issue an indicative or binding economic proposal.	Price, terms, range, conditions and validity.
approve_investment	Authorize the fund to enter the binding investment.	Decision, scope, conditions, vote and authority basis.
execute	Sign or deliver documents.	Instrument, signatory, authority and version.
fund	Release capital.	Amount, account, funds flow, conditions and release actor.

Role, body and threshold controls

Permissions must combine role-based access with transaction-specific authority. A person may hold a title but still lack authority for a particular fund, amount, geography or action.

Control	Implementation
Role permission	Defines which commands a role may request.
Authority grant	Defines which actions the person or body may approve for a fund and threshold.
Quorum and vote rule	Defines attendance, majority, consensus, unanimity, veto or delegated decision.
Conflict status	Removes or limits participation where required.
Delegation	Allows time-bounded delegation with scope and revocation.
Decision validity	Checks correct body, quorum, threshold, scope, expiry and conditions.
Segregation of duties	Separates recommendation, approval, execution and funding where policy requires.
Override permission	Restricts who may deviate from a configurable default.

Do not use a single “approver” flag. The platform must know what action is being authorized and under which fund policy.

Four runtime treatments

AXIOM

A logical, legal or definitional invariant.

- Schema constraint
- State invariant
- Transition guard
- Never configurable

ENFORCE

A required process function.

- Block gate or transition
- Require disposition
- Allow termination instead
- No silent bypass

CONFIGURE

A default controlled by fund or deal conditions.

- Condition-driven default
- Authorized override
- Reason required
- Audit entry

SUGGEST

Helpful practice or conventional artifact.

- Guidance only
- Optional task
- Never blocks by itself
- May be promoted by client policy

The treatment is about product behavior. It is not a statement that every firm uses the same artifact, committee name or operating model.

Axioms and enforced functions

AXIOM AX-01 - Opportunity merits work before material resources are committed

GATE | condition: UNIVERSAL | Opportunity_Assessment -> Material_Resource_Commitment

Implement as a schema constraint, state invariant, or transition guard. It is not client-configurable.

AXIOM AX-02 - A price or offer requires a preliminary economic view

INPUT | condition: WHEN_PRICE_OR_OFFER_IS_ISSUED | Preliminary_Economic_View -> Indicative_or_Binding_Price

Implement as a schema constraint, state invariant, or transition guard. It is not client-configurable.

AXIOM AX-03 - Valid investment authority precedes an unconditional binding commitment

GATE | condition: UNIVERSAL | Valid_Final_Authority -> Binding_Commitment

Implement as a schema constraint, state invariant, or transition guard. It is not client-configurable.

AXIOM AX-04 - Closing conditions and release controls precede funding and closing

GATE | condition: UNIVERSAL | Closing_Readiness -> Funding_and_Closing

Implement as a schema constraint, state invariant, or transition guard. It is not client-configurable.

AXIOM AX-05 - CO_INVESTOR approval precedes CO_INVESTOR commitment and funding

GATE | condition: SPONSOR_LED_COINVESTMENT | CoInvestor_Approval -> CoInvestor_Commitment_and_Funding

Implement as a schema constraint, state invariant, or transition guard. It is not client-configurable.

ENFORCE EMP-01 - Company, seller and management information feeds the model and thesis

INPUT | condition: UNIVERSAL_AT_FUNCTION_LEVEL | Opportunity_Specific_Information -> Model_and_Investment_Thesis

Block advancement until the required process function is satisfied or the transaction is explicitly terminated.

ENFORCE EMP-02 - Sponsor/company information is an input to the co-investor decision

INPUT | condition: SPONSOR_LED_COINVESTMENT | Sponsor_or_Company_Information -> CoInvestor_Final_Decision

Block advancement until the required process function is satisfied or the transaction is explicitly terminated.

ENFORCE EMP-04 - A synthesized and sufficiently bounded case precedes final investment authority

FINALIZATION | condition: UNIVERSAL_AT_FUNCTION_LEVEL | Final_Case_Synthesis -> Final_Investment_Authority

Block advancement until the required process function is satisfied or the transaction is explicitly terminated.

ENFORCE FS-01 - Financial verification feeds normalized earnings, cash flow and valuation

INPUT | condition: UNIVERSAL_AT_FUNCTION_LEVEL | Credible_Financial_Verification -> Normalized_Earnings_Cash_and_Valuation

Block advancement until the required process function is satisfied or the transaction is explicitly terminated.

Configurable rules - part 1

CONFIGURE EMP-03 - Diligence findings revise the model, price, risk register and transaction terms

FEEDBACK | condition: UNIVERSAL_DYNAMIC | Diligence_Findings -> Model_Price_Risk_and_Terms

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE EMP-05 - The acquisition underwrite is converted into ownership accountability and monitoring

INPUT | condition: LEAD_CONTROL_INVESTOR | Underwrite_and_VCP -> Ownership_KPIs_and_Monitoring

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE EMP-06 - Exit assessment and valid disposition authority precede realization

GATE | condition: INVESTMENT_REALIZATION | Exit_Assessment_and_Authority -> Realization

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE FS-02 - Commercial validation feeds material demand, pricing, retention and competition assumptions

INPUT | condition: WHEN_COMMERCIAL_ASSUMPTIONS_ARE_MATERIAL | Commercial_Validation -> Revenue_and_Market_Assumptions

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE FS-03 - Legal, regulatory and tax assessment feeds perimeter, protections and closing requirements

INPUT | condition: WHEN_MATERIAL_TO_TRANSACTION | Legal_Regulatory_and_Tax_Assessment -> Perimeter_Protections_and_Closing_Requirements

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CS-01 - The model and quantified VCP revise one another when value creation drives the underwrite

FEEDBACK | condition: VCP_MATERIAL_TO_RETURNS | Model -> Quantified_Value_Creation_Plan

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

Configurable rules - part 2

CONFIGURE CS-02 - Financing and debt-capacity assessment precedes final approval or signing in leveraged deals

FINALIZATION | condition: `LEVERAGED_ACQUISITION` | `Financing_Assessment -> Final_Approval_or_Signing`

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CS-03 - Carve-out perimeter and standalone work precedes final approval

FINALIZATION | condition: `CORPORATE_CARVEOUT` | `Carveout_Perimeter_TSA_and_Standalone_Work -> Final_Approval`

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CS-04 - Management capability and alignment feed final approval in founder/first-institutional-capital deals

FINALIZATION | condition: `FOUNDER_OWNED_OR_MANAGEMENT_DEPENDENT` | `Management_Talent_Rollover_and_Succession -> Final_Approval`

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CS-05 - A separate authority follows IC recommendation where governance delegates final legal authority

GATE | condition: `DUAL_AUTHORITY_GOVERNANCE` | `IC_Recommendation -> Separate_Board_GP_or_ExCo_Authority`

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CS-06 - Sponsor/company information passes through an intermediary before CO_INVESTOR decision

INPUT | condition: `INTERMEDIATED_COINVESTMENT` | `Sponsor_or_Company_Information -> Intermediary_PPM_then_CO_INVESTOR_Decision`

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CS-07 - Accepted offer or exclusivity precedes confirmatory diligence when access is withheld or broken-deal cost is material

FINALIZATION | condition: `ACCESS_CONSTRAINED_AUCTION_OR_COST_GATED_PROCESS` | `Accepted_Offer_or_Exclusivity -> Confirmatory_Diligence`

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

Configurable rules - part 3

CONFIGURE CS-08 - Incremental economics and integration capacity precede add-on approval

INPUT | condition: BUY_AND_BUILD_ADDON | AddOn_Economics_and_Integration_Capacity -> AddOn_Approval

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CFG-01 - The order of first IC and LOI/term sheet is configurable

GATE | condition: NONE | First_IC -> LOI_or_Term_Sheet

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CFG-02 - External-adviser work can occur before or after LOI/exclusivity

START | condition: NONE | External_Adviser_Work -> LOI_or_Exclusivity

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CFG-03 - Operating-team entry timing is configurable

START | condition: NONE | Operating_Team_Entry -> Screen_or_Post_Conviction

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CFG-04 - The number and names of formal pre-investment gates are configurable

GATE | condition: NONE | PreInvestment_Governance -> Number_and_Names_of_Gates

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CFG-05 - Majority, consensus, supermajority, unanimity and veto rules are configurable

GATE | condition: NONE | Authorized_Decision_Body -> Vote_Threshold_or_Veto

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

Configurable rules - part 4

CONFIGURE CFG-06 - Whether the IC itself is the final legal authority is configurable

GATE | condition: NONE | Investment_Committee -> Final_Legal_Authority

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CFG-07 - Definitive documents may be complete, substantially negotiated or conditionally bounded at final IC

FINALIZATION | condition: NONE | Definitive_Legal_Documents -> Final_IC

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CFG-08 - Exact model-versus-memorandum completion order is configurable and often iterative

FEEDBACK | condition: NONE | Model_Version -> Memorandum_Version

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

CONFIGURE CFG-09 - The depth of co-investor independent diligence is configurable

INFLUENCE | condition: SPONSOR_LED_COINVESTMENT | Sponsor_Underwrite -> Depth_of_CoInvestor_Challenge

Apply the default when its condition is true. Permit an authorized override with a reason and an audit entry.

Suggested practices and optional artifacts

SUGGEST CONV-01 - A formal third-party QoE is an artifact, not the universal function

INFLUENCE | condition: NONE | Formal_ThirdParty_QoE_Report -> Financial_Verification_Function

Surface guidance and optional tasks. Never block a gate solely because this rule is not followed.

SUGGEST CONV-02 - A formal commissioned CDD report is an artifact, not the universal function

INFLUENCE | condition: NONE | Formal_Commissioned_CDD_Report -> Commercial_Validation_Function

Surface guidance and optional tasks. Never block a gate solely because this rule is not followed.

SUGGEST CONV-03 - The 100-day versus 180-day label is conventional

INFLUENCE | condition: NONE | 100_Day_or_180_Day_Label -> Implementation_Readiness

Surface guidance and optional tasks. Never block a gate solely because this rule is not followed.

SUGGEST CONV-04 - Preview/PIM/FIM and similar names are conventional taxonomies

DOCUMENTARY_INHERITANCE | condition: NONE | Named_Memo_Ladder -> Progressive_Decision_Material_Function

Surface guidance and optional tasks. Never block a gate solely because this rule is not followed.

SUGGEST UNV-01 - The same deal team remains involved through ownership or exit

INFLUENCE | condition: NONE | Deal_Team -> Ownership_and_Exit

Surface guidance and optional tasks. Never block a gate solely because this rule is not followed.

SUGGEST UNV-02 - A quantified VCP must always be complete before final approval

FINALIZATION | condition: NONE | PreClose_Quantified_VCP -> Final_Approval

Surface guidance and optional tasks. Never block a gate solely because this rule is not followed.

Configuration flags - transaction structure

Flag	Activate when	System effect
LEVERAGED_ACQUISITION	Debt is material to price, returns or closing certainty.	Require financing assessment, covenant model, liquidity view and funds-certainty controls.
CORPORATE_CARVEOUT	The acquired business depends on a parent or shared services.	Activate perimeter, TSA, standalone cost, migration and separation modules.
FOUNDER_OWNED_OR_MANAGEMENT_DEPENDENT	Continuity, succession or alignment is central.	Increase talent, rollover, governance and transition requirements.
BUY_AND_BUILD_ADDON	The transaction is an add-on or relies on future acquisitions.	Require incremental economics, integration capacity and acquisition headroom.
ACCESS_CONSTRAINED_AUCTION	Confirmatory access follows offer acceptance or exclusivity.	Allow early offer gate with explicit unknowns; require confirmatory diligence before final approval.
INTERMEDIATED_COINVESTMENT	Information passes through an intermediary vehicle or memorandum.	Track information lineage, allocation, intermediary terms and separate co-investor authority.

Configuration flags - governance and economic materiality

Flag	Activate when	System effect
DUAL_AUTHORITY_GOV ERNANCE	One body recommends and another provides final legal authority.	Create two distinct decision events and prevent recommendation from binding the fund.
SPONSOR_LED_COINVE STMENT	A co-investor evaluates an allocation led by another investor.	Activate invitation, information transfer, challenge, approval, allocation and funding flow.
VCP_MATERIAL_TO_RET URNS	Value creation is a significant part of the underwrite.	Require model-VCP feedback, scenario assignment and implementation readiness.
COMMERCIAL_ASSUMPTI ONS_MATERIAL	Demand, pricing, retention or competition drive value.	Require commercial outputs and linked model assumptions.
TRANSACTION_STRUCTU RE_MATERIAL	Legal, tax, regulatory or perimeter issues can change economics or executability.	Require structure, protection and closing-condition modules.
LEAD_CONTROL_INVEST OR	The fund controls governance and ownership execution.	Require ownership budget, KPIs, board governance and value-creation monitoring.
INVESTMENT_REALIZAT ION	A disposal, partial sale or recapitalization is being considered.	Activate exit assessment, disposition authority and realization controls.

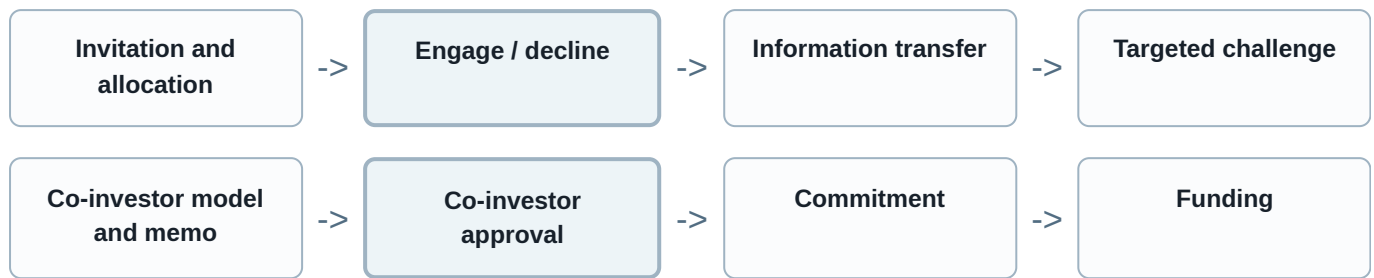
Different firms, same required functions

Organizational design changes who performs a function, how early they enter, and which artifact they produce. It should not delete the underlying function.

Capability	Internal route	External or hybrid route	Kernel behavior
Operating improvement	Portfolio operations team.	Deal team plus specialist adviser or operating partner.	Same operational baseline, initiative and implementation objects.
Capital markets	Internal financing team.	Deal lead, debt adviser, direct lender or arranging bank.	Same debt, covenant, liquidity and certainty fields.
Commercial analysis	Sector team or internal market intelligence.	Commissioned adviser, expert network, direct customer work.	Same commercial assumptions and links.
Financial verification	Internal finance specialists.	Accounting adviser or focused internal review.	Same normalized earnings and cash bridge.
Technology and cyber	Internal technology team.	Specialist adviser, insurer or external assessor.	Same dependency, risk, capex and action objects.
Talent	Internal human-capital team.	Deal team, references, search firm and advisers.	Same role, succession, incentive and hiring objects.
Legal and tax	Internal counsel and tax specialists.	External counsel and tax advisers.	Same issues, protections, conditions and structure objects.

Configuration should route ownership and depth, not create incompatible process templates.

Secondary co-investor workflow



Keep separate

- Lead-investor authority
- Co-investor authority
- Lead assumptions
- Co-investor adjustments
- Allocation and timing constraints
- Separate commitment and funding

Share where appropriate

- Company information
- Transaction terms
- Diligence materials
- Model inputs
- Closing timetable
- Execution documents

The co-investor module is a linked process instance, not a sub-status of the lead-investor process.

Core schemas

Entity	Key fields
FundPolicy	id; strategy; control_profile; mandate_rules; concentration_limits; leverage_policy; authority_policy_id; capability_profile_id
DealProcess	id; fund_id; company_id; transaction_type; phase; status; archetype_flags; lead_owner; materiality_profile; approved_model_id
Workstream	id; deal_id; type; scope; owner_type; owner_id; status; materiality; started_at; closed_for_scope_at
Finding	id; workstream_id; category; conclusion; severity; affected_assumption_ids; affected_open_item_ids; effective_status
ModelVersion	id; deal_id; state; parent_id; version_label; scenario_set_id; checksum; status; decision_record_id
VCPIInitiative	id; deal_id; state; mechanism; baseline; target; value; cost; timing; owner; dependencies; model_links
OpenItem	id; deal_id; category; status; materiality; owner; resolution_path; gate_effect; reopened_from_id
DecisionRecord	id; deal_id; authority_type; body_id; outcome; conditions; scope; valid_until; model_version_id
AuthorityGrant	id; subject_id; authority_type; fund_id; threshold; geography; valid_from; valid_to; delegation_id
ClosingCondition	id; deal_id; condition; owner; status; satisfaction_method; waiver_authority_type; release_effect
ProcessEvent	id; aggregate_type; aggregate_id; event_type; actor_id; payload; timestamp; correlation_id; causation_id

Commands, events and read models

Commands

- AuthorizePursuit
- ApproveOffer
- PublishModelVersion
- ChangeOpenItemStatus
- RecordInvestmentDecision
- ConfirmClosingReadiness
- ReleaseFunds

Events

- PursuitAuthorized
- AssumptionChanged
- ModelPublished
- ItemReopened
- InvestmentApproved
- ConditionSatisfied
- FundsReleased

Read models

- Deal dashboard
- Gate readiness
- Workstream board
- Model bridge
- Open-item register
- Authority log
- Ownership dashboard

Implementation rules

- Commands validate current state, policy, permissions and materiality.
- Events are immutable and idempotent.
- Use correlation and causation identifiers to trace a change across modules.
- Long-running workstreams use eventual consistency; gate commands use strongly consistent checks.
- Read models may be rebuilt from events and authoritative object stores.
- External callbacks must be idempotent and retain the external transaction identifier.

Systems the kernel should connect to

Integration	Direction	Purpose	Control
Deal pipeline / CRM	Bidirectional	Create opportunities, status and ownership.	Kernel remains authoritative for gate state.
Document management / data room	Inbound references and outbound links	Attach materials and versions.	Store references, not uncontrolled copies where policy forbids.
Financial model store	Bidirectional	Publish and load model versions and outputs.	Checksum and immutable approved snapshot.
Task and collaboration tools	Bidirectional	Distribute tasks and comments.	Kernel owns material status and gate effect.
E-signature	Inbound events	Confirm signatures and effective documents.	Verify document version and signatory authority.
Compliance / KYC / AML	Inbound status	Control signing and funding readiness.	Hard block where fund policy requires.
Banking / treasury	Outbound instruction, inbound confirmation	Execute approved funds flow.	Dual control and release authority.
Portfolio monitoring	Bidirectional	Create ownership plan and receive KPIs.	Maintain bridge to approved underwrite.

Primary operating views

Deal cockpit

- Current phase and readiness
- Latest approved and working model
- Material open items
- Decision deadlines
- Critical path and ownership

Workstream board

- Scope and owner
- Tasks and findings
- Assumptions affected
- Open items created
- Local closure status

Decision screen

- Decision version
- Cases and sensitivities
- Material changes
- Conditions and dissent
- Authority validity

Ownership view

- Budget and KPI bridge
- VCP initiatives
- Leverage and liquidity
- Board actions
- Exit readiness

Every screen should answer two questions: “What state is the deal in?” and “What must change before the next authorized action?”

Notifications, escalation and stale-state control

Trigger	System action
Material assumption changed	Recalculate dependent outputs; mark linked readiness stale; notify owners.
Open item overdue	Escalate by materiality and next gate date.
Authority expires	Prevent new action under the expired decision.
Condition deadline approaches	Notify owner, legal and funding roles; expose waiver path.
Model differs materially from approved version	Require comparison and re-approval policy evaluation.
Workstream closed but linked item reopens	Reopen local scope or create a new task.
External document version changes	Invalidate approvals tied to the earlier version where required.
Funding instruction created	Require release controls and segregation of duties.

Avoid notification overload. Only material changes, deadlines, authority actions and blocked transitions should generate mandatory alerts.

Access, audit and transaction integrity

Access model

- Tenant and fund segregation
- Deal-level access groups
- Role and authority separation
- Confidential workstream restrictions
- External adviser access boundaries
- Time-bounded delegation

Audit model

- Immutable decision history
- Assumption and model version history
- Status-transition reasons
- Override and waiver records
- Document-version linkage
- Execution and funding logs

Non-negotiable controls

- No user can approve an action outside the scope of a valid authority grant.
- No approved model can be overwritten.
- No closing condition can be waived without the configured authority.
- No funds can be released from a stale or invalid closing-readiness state.
- No material override can be completed without a recorded reason.
- No external integration can silently change a gate state.

System health and process metrics

Metric	Why it matters
Time in phase	Shows where opportunities stall.
Gate rejection and deferral reasons	Improves screening and resource allocation.
Open-item aging by materiality	Highlights unresolved critical uncertainty.
Model revision frequency and magnitude	Shows volatility and the timing of learning.
Re-approval events	Identifies material change after authority.
Workstream cycle time	Supports capacity planning without equating speed with quality.
Condition satisfaction timing	Improves closing predictability.
Underwrite-to-budget variance	Connects acquisition assumptions to ownership performance.
VCP delivery and realized value	Measures execution against the ownership plan.
Override and waiver rate	Reveals where policy and actual practice diverge.

Metrics should improve operations. They should not create incentives to close workstreams early, minimize recorded uncertainty or avoid legitimate rework.

What the technical lead must prevent

- Building a linear checklist that cannot represent feedback and reopening.
- Using “diligence complete” as a global Boolean.
- Equating a document with the function it is supposed to perform.
- Allowing a model update to replace the approved decision version.
- Using one generic approval permission for spend, offer, investment, execution and funding.
- Hard-coding organizational design into the process kernel.
- Creating separate workflows for different firm sizes instead of routing capabilities.
- Allowing configured defaults to become silent exceptions without a reason.
- Failing to propagate a material finding into model, price, terms, risk or plan.
- Closing an open item without a resolution path and authority.
- Allowing signing or funding from stale readiness.
- Treating co-investor approval as part of the lead-investor authority chain.

The central design test is whether the product can represent institutional complexity and a lean operating model without changing the economic and governance kernel.

Recommended implementation sequence

Stage	Build	Exit criterion
1. Core aggregates	FundPolicy, DealProcess, AuthorityGrant, DecisionRecord, ProcessEvent.	Valid opportunity, pursuit, investment and closing transitions.
2. Readiness and open items	Open-item taxonomy, gate checks, stale-state logic.	Blocked transitions are explainable and auditable.
3. Workstream engine	Seven workstream types, findings, tasks and dependency links.	Material changes can reopen and redirect work.
4. Model and VCP	Model states, assumption graph, scenario controls, VCP maturity.	Decision version and ownership bridge are preserved.
5. Archetype modules	Conditional flags and capability routing.	One kernel supports different deal structures and operating models.
6. Integrations	Documents, models, compliance, signatures, treasury and monitoring.	External events are idempotent and do not bypass controls.
7. Operating views	Cockpit, workstream board, decision screen and ownership dashboard.	Users can understand state, blockers and next authorized action.

Definition of a production-ready kernel

Area	Acceptance criterion
State machine	Invalid authority, commitment and funding sequences are impossible.
Loops	A material finding can update assumptions, reopen items and invalidate readiness.
Workstreams	All required functions can be routed internally, externally or through a hybrid team.
Model control	Working, current and approved versions are distinct and immutable where required.
VCP	Initiatives have mechanisms, value, cost, timing, dependencies, scenario allocation and ownership.
Open items	Every material uncertainty has a state, owner, path, gate effect and reopening history.
Authority	Spend, offer, investment, execution and funding permissions are stored separately.
Configuration	Fund policy and deal flags change defaults without creating a new process spine.
Audit	Every decision, override, waiver, status change and capital release is reconstructable.
Usability	The system clearly shows current state, blocking items, responsible actors and next action.
Resilience	Integrations are idempotent; stale data cannot silently advance a gate.
Ownership continuity	The approved underwrite becomes the starting point for budget, KPIs and VCP monitoring.

The product is complete when it turns investment judgment into controlled, configurable execution without flattening the dynamic nature of underwriting.